Improving Cultural Knowledge Coverage in LLMs via Region-Aware RAG

An Extension of BLEnD — Report: Data, Knowledge Base, Baseline, RAG v1, RAG v2 and a Learned Gate (v3)

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Abstract

We extend the BLEnD benchmark (Myung et al., NeurIPS 2024) with inference-time, region-aware retrieval to test whether retrieval narrows the gap between high- and low-resource cultures. This mid-term report covers the first half of the project: (i) the evaluation subset (5 countries, 9 country–language settings, 4,500 short-answer questions), (ii) an independently built, contamination-safe English cultural knowledge base (720 evidence cards), (iii) a full replication of BLEnD's zero-shot baseline protocol on three free, locally run open-weight LLMs (27,000 model answers, scored with BLEnD's official scorer), (iv) a first Always-RAG run (*RAG v1*, another 27,000 answers) and (v) an improved adaptive pipeline (*RAG v2*). The baseline reproduces the paper's central finding: accuracy falls from 75% (US) to 8% (Ethiopia, in Amharic) on identical questions. RAG v1 (hybrid BM25 + bge-m3, always top-3) *lowers* accuracy by 9–14 points per model, because the KB contains the answer for only 7–14% of questions and small models refuse or get distracted. RAG v2 adds (a) a question-type-driven KB extension for Azerbaijan and Ethiopia (295 new cards), (b) dense retrieval with cross-encoder reranking and a relevance cut-off tuned only on the high-resource countries, (c) a prompt that tells the model to fall back on its own knowledge, and (d) an adaptive gate that keeps the baseline answer when no card is relevant. RAG v2 improves *every* Azerbaijan and Ethiopia setting for all three models (+8.6 to +36.9 points; low-resource mean 28.3→48.7), loses 1–11 points on high-resource countries (old KB), and shrinks the high–low resource gap from 27.2 to 2.0 points. Finally, *RAG v3* learns the gate: a small classifier (gradient-boosted trees, chosen over logistic regression by inner cross-validation) decides per answer whether to keep the RAG or the baseline answer, from retrieval, answer-behaviour and LLM-judge features (a local Mistral-7B rates whether the cards state the answer). Evaluated with template-grouped cross-fitting, so no question is scored by a gate that saw it, v3 recovers almost all of the high-resource loss (50.7→55.2, baseline 55.5), slightly improves the low-resource gain (48.9) and raises the 9-setting mean to 52.4 (baseline 43.4, v2 49.8); a leave-one-country-out evaluation gives 52.1.

1 Scope

BLEnD contains 500 everyday-culture question templates (food, sports, family, education, holidays/celebrations/leisure, work-life). Each is asked about 16 countries, answered by five native annotators per country. We use the **Short-Answer Question (SAQ)** track for **5 countries** that span the resource spectrum. Every non-English country is evaluated twice: with the question in its local language and in English. This gives **9 country–language settings**.

The SOP also lists Assam. It was dropped from this phase because no Assam knowledge base was built. Table 2 breaks the scorable questions down by category. Every setting of a country shares the same scorable set, so the English and local-language versions are directly comparable.

2 Cultural Knowledge Base (KB v1)

For the RAG phase we built one knowledge base (KB) per country. It consists of short English *evidence cards*, each giving one documented cultural fact with its source. To avoid the contami-

Table 1: Countries, languages and question counts. *Scorable* = questions kept by BLEnD's official filter (dropped when ≥3 annotators said "no answer / not applicable" or ≥5 said "I don't know"). Resource class of the language follows Joshi et al. as used in the BLEnD paper.

Country	Local language	Resource	Settings run	Questions / setting	Scorable
United States	English	high	en	500	461
Spain	Spanish	high	es, en	500	469
South Korea	Korean	mid-high	ko, en	500	475
Azerbaijan	Azerbaijani	low	az, en	500	469
Ethiopia	Amharic	low	am, en	500	472
Total: 5 countries, 6 languages			**9 settings**	**4,500**	**4,231**

Table 2: Scorable questions per category (out of the 500 templates: Food 105, Sport 88, Holidays 92, Education 84, Work-life 68, Family 63).

Country	Food	Sport	Holidays	Education	Work-life	Family	Total
US	100	76	81	81	65	58	461
Spain	102	82	78	81	65	61	469
South Korea	102	86	80	82	64	61	475
Azerbaijan	103	82	75	83	67	59	469
Ethiopia	101	77	86	83	67	58	472

nation problem reported in SemEval-2026 Task 7, **no BLEnD material was used**. Sources are institutional: UNESCO intangible-heritage pages, national statistics offices, education and labour ministries, national tourism boards, the US Library of Congress and similar. Every card records its category, topic, region/sub-region, time scope, source URL and a source-quality label.

Table 3: Knowledge-base cards per country and category (target: 25 per category).

Country	Food	Sports	Family	Education	Holidays/Leisure	Work-life	Total
US	25	25	25	25	25	25	150
Spain	25	25	25	25	25	25	150
South Korea	21	19	25	19	25	21	130
Azerbaijan	25	25	25	25	25	25	150
Ethiopia	25	22	25	23	25	20	140
Total							**720**

Retrieval indexes the English text of each card (`text_en`) plus its keywords. For a local-language question, BLEnD's own English version of the question is used *only* as the search query; the LLM still receives the original-language question. The local-language and English versions of a question therefore receive the same cards. A second, question-driven extension of the KB for Azerbaijan and Ethiopia (KB v2) is described in Section 7.1.

3 Baseline: Experimental Setup

Protocol (replicated from BLEnD). Every question is asked with the paper's two prompts, written in the same language as the question:

- `inst-4` (direct): *"Read the following question and provide a single answer without any explanations. Question: {q} Answer:"*
- `pers-3` (persona): *"You are a person from {country} who is trying to explain your country's*

culture to a foreigner. Answer the following question, providing a single answer without any explanations. {q}"

Decoding is greedy (temperature 0), with at most 512 new tokens, as in the paper. No retrieval and no fine-tuning are used. Each model produced 9 settings × 500 × 2 prompts = 9,000 answers, **27,000 in total**.

Models. All three are free, open-weight and run locally. No paid API was used.

Model	Parameters	Format	Licence
Gemma 3 4B-it (`gemma3:4b`)	4.3B	4-bit (Q4_K_M)	Gemma terms
Mistral-7B-Instruct v0.3 (`mistral`)	7.2B	4-bit (Q4_0)	Apache-2.0
Qwen2.5-3B-Instruct (`qwen2.5:3b`)	3.1B	4-bit (Q4_K_M)	Qwen research

Inference ran with Ollama on an institute workstation (NVIDIA RTX 3060, 12 GB). **Deviation from the SOP:** the SOP planned larger models (Qwen3-8B, Aya-101, GPT-OSS-20B, Gemma 2, Llama 3.1) on a 32 GB GPU. The available machine has 12 GB of GPU memory, under 5 GB of free disk, and a firewall that blocks model downloads. We therefore evaluate 3–7B 4-bit models, a *resource-constrained* setting where retrieval has the most room to help.

Scoring (official, unchanged). We use BLEnD's own `soft_exact_match` (`evaluation/exact_match.py`). An answer is correct if any annotator answer appears in it after lemmatisation/stemming. The tools are spaCy (English), Okt/konlpy (Korean), the aznlp stemmer (Azerbaijani) and Spark NLP lemmatisers (Spanish, Amharic). English answer variants are accepted for local-language questions. The released inference and evaluation wrappers do not run as released, so we call the scoring function directly. We report **SEM-B** (binary accuracy, %) averaged over the two prompts, exactly as the paper does.

4 Baseline Results

Table 4: Baseline SAQ accuracy (SEM-B, %), averaged over `inst-4` and `pers-3`. Right block: published scores of two models from the BLEnD paper (Tables 10/11), for reference.

		Ours (4-bit, 3–7B)				**BLEnD paper**	
Country	Lang.	Gemma3-4B	Mistral-7B	Qwen2.5-3B	Mean	Aya-101	Llama-3.1-70B
US	en	72.1	**84.9**	67.8	74.9	53.4	84.9
Spain	es	58.8	**61.3**	48.6	56.2	45.8	75.4
	en	41.2	**65.6**	45.7	50.8	35.7	59.4
South Korea	ko	**54.3**	47.9	41.9	48.0	32.8	65.3
	en	43.7	**60.6**	38.7	47.7	30.7	62.1
Azerbaijan	az	**29.4**	27.7	13.6	23.6	35.8	59.5
	en	45.7	**59.8**	37.1	47.5	31.9	59.5
Ethiopia	am	**14.4**	3.7	5.6	7.9	17.8	17.6
	en	30.6	**46.3**	25.5	34.1	26.4	44.6
Mean, local language		45.8	45.1	35.5	42.1		
Mean, English		46.7	63.4	43.0	51.0		

Table 5: Accuracy (%) by category, pooled over all 9 settings and both prompts.

Model	Food	Sport	Holidays	Education	Work-life	Family
Gemma3-4B	38.8	43.8	36.2	49.1	52.9	41.3
Mistral-7B	46.8	52.5	46.1	55.8	57.1	47.5
Qwen2.5-3B	30.7	33.4	30.9	43.8	44.2	35.5

4.1 Observations

1. **The cultural gap is large and reproduces the paper.** In the local language, the 3-model mean falls from 74.9% (US) to 56.2% (Spain), 48.0% (South Korea), 23.6% (Azerbaijan) and 7.9% (Ethiopia). The paper's 16-model averages show the same ordering (US 79.2, Spain 69.1, Ethiopia 12.2). High-resource local settings average 53–65%; low-resource ones average 10–22%.
2. **Low-resource cultures are answered better in English.** Azerbaijan improves from 23.6% to 47.5% and Ethiopia from 7.9% to 34.1% when the same question is asked in English. For Spanish and Korean the local language is as good or better (e.g. Gemma: 54.3% Korean vs 43.7% English). This matches BLEnD's finding exactly.
3. **Amharic is nearly unsolved by small models** (3.7–14.4%). Many outputs are not fluent Amharic, e.g. repetitions of the prompt.
4. **Our models sit between the paper's Aya-101 and Llama-3.1-70B.** Mistral-7B matches Llama-3.1-70B on US and English settings. All our models remain well below it on non-English local-language settings.
5. **Category effects match the paper:** food and holidays are hardest, while education and work-life are easiest, for all three models (Table 5).

4.2 Caveats of the official metric

- **Verbosity.** Mistral ignores "single answer" more often: median answer length is 13 words, against 2 for Gemma and Qwen. Since the metric checks whether a gold answer is *contained* in the response, longer answers have more chances to match. Part of Mistral's lead may come from this.
- **Language switching under the persona prompt.** With `pers-3`, Gemma often answers English questions about Spain in Spanish ("Naranja" for "orange"). English settings accept only English gold answers, so its Spain-English score drops from 48.4% (`inst-4`) to 33.9% (`pers-3`).
- **Morphologically rich languages.** Lemma/stem matching under-counts correct answers in morphologically rich languages, as noted in the SOP.
- **Software versions.** Two scorer libraries are newer than the paper's because of Python/Java compatibility: Spark NLP 5.5.3 (paper: 5.3.3) and pandas 2.2.3 (2.2.2). The lemmatiser models and the scoring code are unchanged.

5 RAG v1: Always-RAG

Retrieval. Retrieval is region-aware hybrid search over the target country's KB only (SOP §III-A). Each card (`text_en` + keywords) is scored by BM25Okapi (`rank_bm25`, k_1=1.5, b=0.75) and by the cosine similarity of `BAAI/bge-m3` embeddings. Both score lists are min-max normalised per query and combined as $0.5 \cdot \text{dense} + 0.5 \cdot \text{BM25}$. The top 3 cards are kept. The query is always BLEnD's English version of the question, so the local-language and English versions of a question receive the same cards. All settings were fixed before seeing any RAG results.

Generation. The prompt is the *unchanged* baseline prompt (`inst-4` / `pers-3`, in the question's language), preceded by *"Background information about {country} (may or may not be relevant):*

[1] ... [2] ... [3] ...”. Models, decoding and scoring are identical to the baseline: another 27,000 answers.

Retrieval coverage. We use the gold answers only as a diagnostic. A gold English answer appears in the retrieved top 3 cards for 13.0% (US), 14.3% (Spain), 6.7% (South Korea), 9.8% (Azerbaijan) and 11.0% (Ethiopia) of scorable questions.

Table 6: Always-RAG vs baseline (SEM-B %, mean of the two prompts): baseline → RAG, and the change Δ. Positive changes in bold.

		Baseline → RAG			Δ (RAG − baseline)			
Country	Lang.	Gemma3-4B	Mistral-7B	Qwen2.5-3B	Gemma	Mistral	Qwen	Mean Δ
US	en	72.1→46.0	84.9→65.7	67.8→48.7	−26.1	−19.2	−19.1	−21.5
Spain	es	58.8→33.0	61.3→44.8	48.6→32.5	−25.8	−16.5	−16.1	−19.5
	en	41.2→28.4	65.6→46.2	45.7→28.4	−12.8	−19.4	−17.3	−16.5
South Korea	ko	54.3→40.3	47.9→42.4	41.9→34.8	−14.0	−5.5	−7.1	−8.9
	en	43.7→30.0	60.6→49.3	38.7→28.1	−13.7	−11.3	−10.6	−11.9
Azerbaijan	az	29.4→20.5	27.7→18.3	13.6→15.1	−8.9	−9.4	**+1.5**	−5.6
	en	45.7→27.1	59.8→40.7	37.1→24.1	−18.6	−19.1	−13.0	−16.9
Ethiopia	am	14.4→16.4	3.7→5.3	5.6→15.3	**+2.0**	**+1.6**	**+9.7**	**+4.4**
	en	30.6→20.2	46.3→33.2	25.5→17.8	−10.4	−13.1	−7.7	−10.4
All 9 settings		43.4→29.1	50.9→38.4	36.1→27.2	−14.3	−12.4	−8.9	−11.9

5.1 Findings

1. **Always-RAG hurts overall** (−8.9 to −14.3 points per model). It improves only 3 of 27 model–setting cells, all in the two lowest-resource local-language settings: Amharic for all three models (3-model mean 7.9→12.3; Qwen +9.7) and Azerbaijani for Qwen (+1.5).
2. **The more the model already knows, the more retrieval hurts.** US (−21.5) and Spain (−19.5) lose most; South Korea and Azerbaijani lose 6–12; Amharic, with a 4–14% baseline, gains. This is the pattern Team CultRAG reported on this benchmark, and it motivates the adaptive gate (PS-2).
3. **Failure modes.** Across all model answers, 4,934 changed from correct to wrong and 1,932 from wrong to correct. Inspection shows two failure modes: (i) *refusal*: 7–20% of RAG answers per country (vs. ≤0.4% at baseline) say the provided background does not contain the answer, even when the model answers correctly without it; (ii) *distraction*: the model answers with something mentioned in a loosely related card. Refusals account for 22% of the correct→wrong flips. Replacing every refusal with the model's baseline answer still leaves RAG 7–11 points below the baseline, so distraction is the dominant cause.
4. **Root cause: KB coverage.** With a gold answer in the top 3 cards for only 7–14% of questions, most retrieved context is noise for these everyday-life questions. That is the kind of knowledge BLEnD notes is rarely written down in institutional sources.

6 Diagnosis of RAG v1 and Related Work

RAG works when the right card is retrieved. Splitting the RAG v1 answers by whether a gold-mentioning card was in the top 3 (diagnostic only) shows that retrieval itself is useful, but rarely succeeds (Table 7). When the right card is present, accuracy rises sharply, most in the local languages (Azerbaijani 26→51, Amharic 8→30). When it is absent (86–89% of questions), the cards are noise and accuracy falls.

Table 7: RAG v1 accuracy (%, all models and prompts pooled) split by whether a gold answer appears in the 3 retrieved cards. "Base" is the baseline accuracy on the same questions.

		Gold in top-3 (11–14%)		Gold not in top-3	
Country	Lang.	Base	RAG v1	Base	RAG v1
Azerbaijan	az	26.4	**50.6**	23.2	13.8
	en	62.6	**70.4**	45.6	25.6
Ethiopia	am	7.6	**30.1**	8.0	9.3
	en	44.6	**49.5**	32.4	19.4
All 9 settings		49.6	**56.1**	42.5	28.0

Fusion weights. Re-scoring the v1 retrieval with different BM25/dense weights (Table 8) shows that BM25 *hurts*: questions share generic words ("most popular", the country name) with many cards. Dense-only retrieval finds a gold-mentioning card most often for every country.

Table 8: Answer-hit@3 (%) of KB v1 for different fusion weights (α = weight of the dense score; RRF = reciprocal-rank fusion with k=60).

Country	BM25 only	α=0.3	α=0.5 (v1)	α=0.7	Dense only	RRF
US	8.2	12.1	13.0	15.8	**18.4**	13.0
Spain	9.8	11.9	14.3	16.2	**19.2**	14.1
South Korea	3.8	5.1	6.7	10.1	**15.6**	8.2
Azerbaijan	6.4	9.0	9.8	11.3	**14.1**	10.4
Ethiopia	7.2	8.3	11.0	16.1	**17.4**	14.2

SemEval-2026 Task 7. BLEnD was the basis of SemEval-2026 Task 7 (62 teams). The lessons we take from it:

- **king001** (1st place, SAQ) used embedding retrieval with GPT-5.2. Their KB was the official BLEnD question–answer data itself, which explains their 98–99% on Amharic and Azerbaijani; we do not copy this, since it amounts to answer leakage. We do adopt two of their ideas: use *no* context when nothing relevant is found, and keep answers short.
- **CultRAG** (hybrid BM25 + dense + RRF) found that plain RAG did not beat the LLM-only baseline, because only 40.7% of questions were answerable from their KB. RAG helped under-represented cultures (e.g. Ethiopia +4.1) and hurt the US/UK. The only component that helped was a prompt against anchoring on the context, and they recommend selective retrieval and targeted KB expansion.
- **Simorgh** used $0.4 \cdot \text{BM25} + 0.6 \cdot \text{dense}$ with a regional boost, top-5, without ablations.

Our diagnosis agrees with CultRAG: KB coverage is the bottleneck, and context must be gated.

7 RAG v2: Question-Driven KB + Reranked, Gated Retrieval

RAG v2 changes four things. Models, decoding, prompts' question part and scoring are identical to the baseline.

7.1 KB v2: question-driven extension (Azerbaijan, Ethiopia)

KB v1 cards are institutional and often niche (e.g. a Sidama festival dish), while BLEnD asks about mainstream everyday life (most popular sport, typical breakfast, school starting age). We therefore wrote new cards for the two low-resource countries, driven by the BLEnD *question types*: for each template we wrote a card stating the mainstream fact in the style "The most common X in Ethiopia

is Y; also common are Z", with local-language terms in brackets (e.g. the Amharic-script spelling of *injera*, or Azerbaijani *çay* for tea). Facts come from cited sources: national labour law (Ethiopia Labour Proclamation 1156/2019, Azerbaijan Labour Code summaries), Wikipedia articles on each country's cuisine, education, public holidays, sport, culture and economy, and news reports. Each card stores the question IDs whose template motivated it (`written_for`). The cards are kept in a separate file (`KB_Articles/<Country>_KB_v2_additions.jsonl`); retrieval searches the union of KB v1 and KB v2 as one pool, and `written_for` is never used at retrieval time.

Table 9: KB v2 cards per category, and question templates covered (out of 500).

Country	Food	Sports	Family	Education	Holidays	Work-life	Total	Templates
Azerbaijan	36	19	29	27	16	19	146	440
Ethiopia	39	17	27	25	21	20	149	432

Coverage. Answer-hit@3 with dense retrieval rises from 17.4% to 57.4% (Ethiopia) and from 14.1% to 63.5% (Azerbaijan). To check that the cards hold general knowledge rather than one answer per question, we also retrieved with each question's own `written_for` cards removed: hit@3 is still 37.9% (Ethiopia) and 35.2% (Azerbaijan), about 2.5× KB v1.

Honesty statement. Gold answers were not consulted while writing cards, but the KB was written *after* the RAG v1 error analysis, during which the 25 most frequent gold answers per country were seen. KB v2 is therefore question-aware, not blind. About one third of the cards (53 for Ethiopia, 46 for Azerbaijan) are directly backed by a source we checked ("high"); the rest state widely known everyday facts and cite a general reference page ("medium"). US, Spain and South Korea keep KB v1 only.

7.2 Retrieval and gating

1. **Candidates:** $\alpha \cdot \text{dense} + (1-\alpha) \cdot \text{BM25}$ (both min-max normalised per query) over the country's KB; top 20 candidates. α is chosen from $\{0.8, 1.0\}$ by recall@20 on the development countries: 33.5% vs 34.0%, so α=1.0 (dense only).
2. **Reranking:** each (English question, card) pair is scored by the cross-encoder `cross-encoder/ms-marco-Mini` giving a relevance probability in $[0, 1]$. We first tried the multilingual `BAAI/bge-reranker-v2-m3`, but it ran at about 1 pair/s on the laptop CPU (≈12 h for 50,000 pairs). Queries and cards are English, so the small English reranker is adequate; it scored all pairs in about 20 minutes.
3. **Cut-off:** keep at most K=3 cards with relevance $\geq \tau$. A question therefore gets 0–3 cards.
4. **Tuning τ without the test countries:** US, Spain and South Korea are the development set. With "card mentions a gold answer" as the relevance label, τ maximises the precision-weighted $F_{0.5}$ over the cards that would be sent, on a grid $0.05, 0.10, \ldots, 0.95$. The best value is τ=0.45 ($F_{0.5}$=0.129). It was then frozen for Azerbaijan and Ethiopia, whose gold answers were never used for tuning.

7.3 Prompt v2 and the adaptive gate

If at least one card passes the cut-off, the unchanged baseline prompt is preceded by: *"Background information about {country}. Use it only if it directly helps to answer the question; otherwise ignore it and answer from your own knowledge. Always give an answer and never say that the information is not provided. [1] ..."* This targets the two v1 failure modes, refusal and distraction. If no card passes, the prompt is byte-identical to the baseline prompt. Decoding is greedy, so the answer equals the stored baseline answer, which is reused without a new model call. This is the training-free Adaptive-RAG gate of the SOP (PS-2).

Table 10: RAG v2 configuration (all values fixed before evaluating Azerbaijan and Ethiopia).

Component	Setting
Retrieval query	BLEnD's English question (also for local-language questions)
Search scope	target country's KB only (v1 ∪ v2 for AZ/ET; v1 for US/ES/KR)
Dense encoder	`BAAI/bge-m3`, cosine similarity, max length 256
BM25	`rank_bm25` BM25Okapi, k_1=1.5, b=0.75, sklearn English stop-words
Fusion weight α (dense)	1.0 (chosen on dev from $\{0.8, 1.0\}$)
Candidates	top 20
Reranker	`cross-encoder/ms-marco-MiniLM-L-12-v2`, max length 256
Relevance cut-off τ	0.45 (max $F_{0.5}$ on dev)
Max cards K	3
No relevant card	baseline prompt ⇒ baseline answer reused
Prompt	v2 instruction (above) + unchanged `inst-4/pers-3`
Models / decoding	as baseline (greedy, temperature 0, seed 0, max 512 tokens)

Table 11: RAG v2 retrieval statistics per country (scorable questions for the last two columns).

Country	KB cards	Questions with 0 / 1 / 2 / 3 cards				Answer-hit
US	150	467	27	6	0	1.5%
Spain	150	289	115	57	39	8.7%
South Korea	130	288	117	55	40	5.5%
Azerbaijan	296	1	5	18	476	64.6%
Ethiopia	289	1	7	10	482	58.9%

For Azerbaijan and Ethiopia, 95–97% of the cards sent come from KB v2. For the high-resource countries the cut-off withholds context for 58–93% of questions. In total, 3,992 (AZ/ET) + 1,758 (US/ES/KR) prompts per model received context and were run; the rest reuse the baseline answers.

7.4 Results

7.5 Findings

1. **RAG v2 improves every low-resource setting for every model**, by +8.6 to +36.9 points (Table 12). The low-resource mean rises from 28.3% to 48.7%. The largest gains are in the local languages (Azerbaijani +29.7, Amharic +20.1), the settings the SOP targets.
2. **The resource gap nearly closes.** The high–low gap falls from 27.2 points (baseline) to 2.0 (Table 13). RAG v1 also narrowed the gap, but only by making everything worse.
3. **High-resource countries still lose a little** (−1.7 to −6.4 points mean, versus −8.8 to −21.5 for v1). They still use the niche KB v1, so the few cards that pass the cut-off are sometimes misleading. The US is almost unaffected because 93% of its questions get no context.
4. **The failure modes of v1 are largely fixed.** Refusal-like answers drop from 7–20% (v1) to 1–5% (v2) per country. For Azerbaijan and Ethiopia, 3,017 answers changed from wrong to correct and 716 from correct to wrong (v1: 810 vs 1,614).
5. **What drives the gain.** Tables 7 and 11 together show that the gain comes mainly from KB coverage: the same retrieval and prompt ideas cannot help when the KB does not contain everyday knowledge. The gate and prompt v2 limit the damage when it does not.

7.6 Limitations of RAG v2

- KB v2 is question-aware (written from BLEnD's question templates after error analysis), so its gains are an optimistic estimate for unseen questions; the leave-own-cards-out coverage (35–38%

Table 12: All three systems (SEM-B %, mean of `inst-4` and `pers-3`): Baseline (B), RAG v1 and RAG v2, and the 3-model mean change of each RAG version. US, Spain and South Korea are the development countries for α and τ and use KB v1 only; Azerbaijan and Ethiopia are held out.

		Gemma3-4B			Mistral-7B			Qwen2.5-3B			Mean Δ	
Country	Lang.	B	v1	v2	B	v1	v2	B	v1	v2	v1	v2
US	en	72.1	46.0	70.8	84.9	65.7	83.4	67.8	48.7	65.4	−21.5	−1.7
Spain	es	58.8	33.0	53.4	61.3	44.8	56.0	48.6	32.5	40.5	−19.5	−6.3
	en	41.2	28.4	40.0	65.6	46.2	58.5	45.7	28.4	34.9	−16.5	−6.4
South Korea	ko	54.3	40.3	48.4	47.9	42.4	45.5	41.9	34.8	37.3	−8.8	−4.3
	en	43.7	30.0	39.8	60.6	49.3	55.8	38.7	28.1	31.4	−11.9	−5.4
Azerbaijan	az	29.4	20.5	**56.1**	27.7	18.3	**53.3**	13.6	15.1	**50.5**	−5.6	**+29.7**
	en	45.7	27.1	**63.6**	59.8	40.7	**71.4**	37.1	24.1	**61.2**	−16.9	**+17.9**
Ethiopia	am	14.4	16.4	**40.7**	3.7	5.3	**17.2**	5.6	15.3	**26.2**	+4.4	**+20.1**
	en	30.6	20.2	**46.5**	46.3	33.2	**54.9**	25.5	17.8	**42.7**	−10.4	**+13.9**
All 9 settings		43.4	29.1	**51.0**	50.9	38.4	**55.1**	36.1	27.2	**43.3**	−11.9	**+6.4**

Table 13: Summary: mean accuracy over high-resource settings (US, Spain, South Korea; 5 settings) and low-resource settings (Azerbaijan, Ethiopia; 4 settings), and the gap between them.

	High-resource (per model)			Low-resource (per model)			
System	Gemma	Mistral	Qwen	Gemma	Mistral	Qwen	Gap (mean)
Baseline	54.0	64.1	48.6	30.0	34.4	20.5	27.2
RAG v1	35.5	49.7	34.5	21.1	24.4	18.1	18.7
RAG v2	50.5	59.8	41.9	**51.7**	**49.2**	**45.1**	**2.0**

hit@3) is the more conservative figure.

- Two thirds of KB v2 cards are not individually source-verified.
- The cut-off was tuned on countries that have only KB v1, so it is untuned for the denser KB v2 (99.8% of AZ/ET questions receive context).
- Answer-hit is a lexical proxy for relevance, used only for tuning on dev countries and for diagnostics.

8 RAG v3: a Learned Gate

8.1 Motivation: the gate of v2 is too coarse

RAG v2 decides *per question* with one rule: send the cards if any reranker score is $\geq \tau$. Comparing, for every answer that received cards, the RAG v2 answer with the baseline answer of the same model and prompt shows where this rule fails (Table 14). In the high-resource countries RAG hurts two to three times more often than it helps, yet a *perfect* per-answer choice between the two existing answers would beat the baseline in every setting. The headroom is large, so the question is whether the choice can be learned.

8.2 The journey in short

1. **Reranker score is not usefulness.** In the high-resource countries even cards with reranker score ≥ 0.99 hurt on average (mean change -0.21 per answer). The cross-encoder measures *topic* relevance; KB v1 cards are on-topic institutional text that rarely contains the everyday answer. A stricter τ cannot fix this.
2. **A simple confidence rule does not work either.** Using RAG only when the model's two baseline prompts disagree (a self-consistency signal) leaves the high-resource change negative in

Table 14: Answers that received cards in RAG v2 (all 3 models, both prompts): how often the cards turned a wrong baseline answer right (help) or a right one wrong (hurt), and SEM-B on this subset for the baseline, RAG v2 and a perfect gate that always keeps the better of the two answers.

Country	Lang.	Help	Hurt	Baseline	RAG v2	Perfect gate
US	en	7	55	82.3	57.3	85.9
Spain	es	120	297	60.3	45.8	70.1
	en	129	308	55.4	40.8	65.9
South Korea	ko	140	263	49.9	39.9	61.3
	en	107	260	46.9	34.5	55.6
Azerbaijan	az	977	141	23.6	53.4	58.4
	en	760	257	47.5	65.4	74.5
Ethiopia	am	650	81	7.9	28.1	30.9
	en	630	237	34.2	48.1	56.5

both groups.

3. **Literature.** Adaptive retrieval is a known problem: retrieval helps mainly for long-tail knowledge (Mallen et al., ACL 2023); SKR (Wang et al., EMNLP Findings 2023) trains a small classifier to retrieve only when the model does not already know; CRAG (Yan et al., 2024) scores retrieved documents with a light evaluator; Adaptive-RAG (Jeong et al., 2024) routes by query complexity. We follow SKR/CRAG: a small supervised gate trained on *outcome* labels.
4. **Problem: training on the same countries we test on.** The gate matters most for US, Spain and South Korea, so they cannot simply be the training set. We use *cross-fitting*: every answer is scored by a gate that never saw its question (Section 8.4).
5. **First gate, then a judge.** A gate on retrieval and answer-behaviour features already recovered most of the loss, but could hardly separate good from bad cards *within* Spain and South Korea. We therefore added an LLM judge that answers the narrower question "do these cards state the answer?" and used its probability as one more feature.

8.3 Task, features and labels

The gate is **answer selection**: for every answer that received cards, both the baseline answer and the RAG v2 answer already exist (and are officially scored), so the gate only picks one; no new LLM calls are needed. The label is *help* = 1 if the RAG answer is correct and the baseline answer is wrong, 0 for the reverse; ties do not change accuracy and are not used for training. This gives 17,250 rows with cards, of which 5,419 are decisive (3,520 help, 1,899 hurt). The features use only information available at inference time, never the gold answers (Table 15). Country names are deliberately *not* a feature, so the gate can be applied to a new country.

Table 15: Gate features (one row per model, country, language, prompt and question).

Group	Features
Retrieval	`r_max`, `r_mean`, `r_min` (reranker scores of the cards that passed τ), `r_gap` (best minus second best), `n_cards` (1–3)
Behaviour	`base_agree`: baseline `inst-4` and `pers-3` answers identical (self-consistency); `base_in_cards`, `rag_in_cards`: share of the answer's words that occur in the cards or their local-language keywords; `changed`: cards changed the answer; `rag_base_overlap`; `base_len`, `rag_len` (words)
Context	model, prompt (`inst-4`/`pers-3`), English vs local language, topic (6)
Judge	`judge_yes`: P(YES) from Mistral-7B (Ollama, run locally, greedy, read from the first token's log-probabilities) for "Do the background notes explicitly state an answer to this question (not just mention the topic)?". Input: the English question and the cards only; one call per question with cards (1,454 calls, shared by all models, languages and prompts)

What the judge says. It finds the answer in only 6–7% of the card sets for the US, Spain and South Korea, but in 43–47% for Azerbaijan and Ethiopia (share with $P(\text{YES}) > 0.5$). This independently confirms the diagnosis of Table 7: KB v1 cards are on topic but rarely contain the everyday answer.

8.4 Training protocol (no question is scored by a gate that saw it)

- **Grouped 5-fold cross-fitting.** BLEnD asks the same question *template* in every country, language, prompt and model, so a random split would leak. Folds are grouped by template (`GroupKFold`): a template is either entirely in training or entirely in the test fold. Each fold's predictions come from a gate trained on the other four; all five countries are thus tested out-of-sample.
- **Leave-one-country-out (LOCO).** Stricter: to predict country C, train only on the other four countries *and* only on other templates. This simulates adding a new country.
- **Model and threshold selection** happen inside each training split with an inner 3-fold grouped CV: logistic regression (linear, standardised features, balanced classes) versus gradient-boosted trees (`HistGradientBoosting`, depth 3, 200 iterations, balanced classes), chosen by AUC, and the probability threshold $t \in \{0.30, 0.35, \ldots, 0.70\}$ that maximises the number of answers gained over the baseline. The RAG answer is used iff $P(\text{help}) > t$.
- Three feature sets are compared: retrieval only; + behaviour; + behaviour + judge (all).
- Gradient boosting was selected in every outer fold of every setup (5/5 and 25/25); with all features the chosen thresholds were 0.30–0.35 for cross-fitting and 0.30–0.45 for LOCO.

Table 16: Gate quality: out-of-fold ROC-AUC for predicting help vs hurt, overall and within each country, for the three feature sets (R = retrieval, +B = + behaviour, +J = + behaviour + judge).

	Grouped 5-fold			Leave-one-country-out		
	R	+B	+J	R	+B	+J
All rows	0.797	0.822	**0.824**	0.790	0.813	0.813
US	0.584	0.751	0.738	0.639	0.771	**0.800**
Spain	0.584	0.609	**0.644**	0.546	0.552	**0.594**
South Korea	0.574	0.596	**0.622**	0.555	0.598	**0.607**
Azerbaijan	0.686	0.744	0.736	0.671	0.722	0.710
Ethiopia	0.682	0.746	0.750	0.656	0.717	0.725

8.5 What the gate learned

Logistic-regression weights on standardised features (fitted on all decisive rows, for interpretation only) give a readable picture of the decision; positive means "use the RAG answer":

- **The judge is the strongest single signal** (`judge_yes` +0.38): if an LLM says the cards state the answer, use the RAG answer.
- **Trust the cards** when even the weakest is relevant (`r_min` +0.37), there are more of them (`n_cards` +0.34) and the RAG answer is taken from them (`rag_in_cards` +0.28). This is the profile of KB v2 cards.
- **Keep the baseline** when the baseline answer already appears in the cards (`base_in_cards` −0.33: the cards add nothing), when one card dominates the others (`r_gap` −0.30), when the model is self-consistent without cards (`base_agree` −0.18), and for sport questions (−0.38).
- **By model**, Mistral-7B, the strongest model, benefits least from cards (−0.17) and Qwen2.5-3B, the weakest, most (+0.14): a model-level version of the "retrieve only when the model does not know" idea. English questions benefit less than local-language ones (−0.22), consistent with BLEnD's finding that models know more about low-resource cultures in English.

Table 17: Baseline (B), RAG v2 and RAG v3 (learned gate with all features incl. the judge, grouped 5-fold cross-fitting) per model, SEM-B % (mean of `inst-4` and `pers-3`), and the 3-model mean change versus the baseline.

		Gemma3-4B			Mistral-7B			Qwen2.5-3B			Mean Δ	
Country	Lang.	B	v2	v3	B	v2	v3	B	v2	v3	v2	v3
US	en	72.1	70.8	72.2	84.9	83.4	84.9	67.8	65.4	67.8	−1.7	**0.0**
Spain	es	58.8	53.4	58.4	61.3	56.0	60.8	48.6	40.5	47.8	−6.3	**−0.6**
	en	41.2	40.0	41.4	65.6	58.5	65.4	45.7	34.9	44.7	−6.4	**−0.4**
South Korea	ko	54.3	48.4	53.5	47.9	45.5	48.3	41.9	37.3	41.1	−4.3	**−0.4**
	en	43.7	39.8	42.8	60.6	55.8	60.1	38.7	31.4	38.9	−5.4	**−0.4**
Azerbaijan	az	29.4	56.1	55.8	27.7	53.3	52.9	13.6	50.5	49.3	+29.7	+29.0
	en	45.7	63.6	64.0	59.8	71.4	71.5	37.1	61.2	62.3	+17.9	**+18.4**
Ethiopia	am	14.4	40.7	41.4	3.7	17.2	16.9	5.6	26.2	25.8	+20.1	**+20.2**
	en	30.6	46.5	47.8	46.3	54.9	55.4	25.5	42.7	43.5	+13.9	**+14.8**
All 9 settings		43.4	51.0	**53.0**	50.9	55.1	**57.4**	36.1	43.3	**46.8**	+6.4	**+9.0**

Table 18: Summary over all systems (3-model means, SEM-B %). v3 rows are out-of-sample; the perfect gate is an upper bound that uses the gold labels. LOCO = leave-one-country-out.

System	High-resource	Low-resource	All 9	Gap
Baseline	55.5	28.3	43.4	27.2
RAG v1	39.9	21.2	31.6	18.7
RAG v2	50.7	**48.7**	49.8	2.0
v3: retrieval (5-fold)	55.0	48.2	52.0	6.8
v3: + behaviour (5-fold)	55.1	48.5	52.2	6.6
v3: + behaviour + judge (5-fold)	**55.2**	**48.9**	**52.4**	6.3
v3: + behaviour (LOCO)	55.0	48.1	52.0	6.9
v3: + behaviour + judge (LOCO)	55.1	48.4	52.1	6.7
Perfect gate (upper bound)	59.1	55.0	57.3	4.1

8.6 Results

8.7 Findings

1. **The learned gate removes almost all of v2's high-resource loss** (high-resource mean 50.7→55.2, baseline 55.5) and even slightly improves the low-resource result (48.7→48.9). The 9-setting mean improves from 49.8 (v2) to **52.4**, i.e. +9.0 points over the baseline, and every model improves over v2 (Table 17).
2. **It transfers to unseen countries.** Leave-one-country-out loses only 0.3 points (52.1), so the gate is not memorising country identity.
3. **Of the 5,419 decisive answers**, the gate avoided 1,129 of the 1,899 harmful switches at the cost of 476 of the 3,520 helpful ones; it keeps the baseline answer for 33% of the answers that received cards.
4. **The judge helps where it was meant to, but modestly.** It is the strongest single feature and raises the within-country AUC for Spain (0.61→0.64) and South Korea (0.60→0.62), yet adds only +0.2 points to the 9-setting mean (+0.1 under leave-one-country-out). Most of the gain comes from the retrieval and behaviour features.
5. **Where it is still weak.** Within Spain and South Korea the AUC stays near 0.6, so these settings remain 0.4–0.6 points below the baseline, and the gap widens again to 6.3 points because the high-resource side recovers. The perfect gate (57.3) shows another 5 points of headroom.

8.8 Limitations of RAG v3

- The labels come from BLEnD's gold answers. Cross-fitting keeps every reported number out-of-sample, but the gate is still tuned to this benchmark and these three models.
- It selects between two existing answers, so at inference time it needs both the baseline and the RAG answer plus one judge call (three LLM calls for questions with cards). On a laptop CPU the judge took about 12 s per question.
- The judge is a 7B model reading English cards; its probabilities are very confident (81% are below 0.05 or above 0.95), which limits how finely the gate can use them.
- The behaviour features are lexical (word overlap) and work less well for Amharic, whose answers rarely share words with English cards.

9 Next Steps

1. **Leave-own-cards-out run:** re-run Azerbaijan and Ethiopia with each question's own KB v2 cards removed, to measure accuracy (not only coverage) on effectively unseen questions.
2. **Oracle-KB upper bound:** RAG with cards built from the BLEnD answers, run separately and labelled as an upper bound, to separate the effect of knowledge coverage from the RAG method.
3. **KB v2 for US, Spain and South Korea**, to test whether question-driven cards remove the remaining high-resource losses.
4. **Cross-lingual retrieval:** machine-translated English queries for Azerbaijani and Amharic questions, compared against the oracle English query.
5. **Category-aware analysis** of where retrieval helps or hurts, from the per-question logs.

Reproducibility. The code (`rag_pipeline/`), the KBs, all prompts sent, all model responses (baseline, RAG v1, RAG v2), the gate features, judge outputs and predictions (RAG v3) and per-question scores are in the project repository (`https://github.com/SidRai-247/BlenD_ML_Project`).